higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T140**(E)**(A12)T

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL CONSTRUCTION N5

(8060015)

12 April 2019 (X-Paper)
09:00–13:00

OPEN-BOOK EXAMINATION

REQUIREMENTS: **ONE A2 drawing sheet**
Hot-rolled structural steel sections (BOE 8/2)

This question paper consists of 7 pages and 1 formula sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
BUILDING AND STRUCTURAL CONSTRUCTION N5
TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Draw a 20 mm border line around the DRAWING SHEET on both sides.
5. Use only pencil on the DRAWING SHEET.
6. Use both sides of the DRAWING SHEET.
7. Draw ALL drawings according to the required scale.
8. Round off calculations to THREE decimal places.
9. Use your discretion where dimensions are NOT given.
10. Write neatly and legibly.

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SECTION A

QUESTION 1: LOADED BEAM

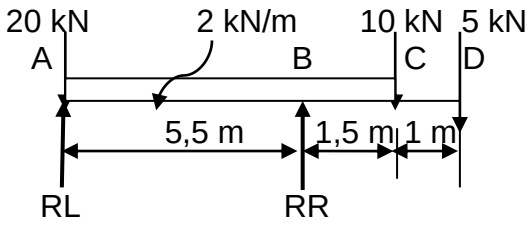


FIGURE 1

FIGURE 1 shows a simple supported beam.

1.1 Calculate the reactions RL and RR. (4)

1.2 Calculate the shear-force values and draw to a suitable scale a shear-force diagram. (8)

1.3 Calculate the maximum bending-moment values and draw to a suitable scale a bending-moment diagram. (4)

[16]

QUESTION 2: SELECT STEEL BEAM

The maximum bending moment and maximum shear force of a loaded simple supported beam are 29,75 kN/m and 18 kN respectively. Select from the hot-rolled structural steel sections (BOE 8/2) a suitable I-section with tapered flange if the maximum bending stress may NOT exceed 165 MPa.

2.1 Calculate the section modules. (4)

2.2 Investigate the safety factors. (4)

[8]

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QUESTION 3: CENTROIDS

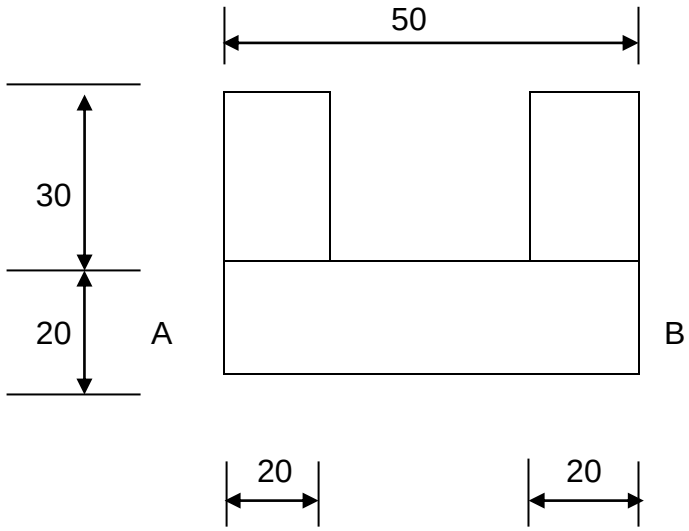


FIGURE 2

FIGURE 2 shows a cross section through a steel section channel. ALL dimensions are given in mm.

Calculate:

3.1 The distance of the neutral X-X (A-B) axis (4)

3.2 The moment of inertia of the plate about the neutral X-X axis (5)

3.3 The profile modules (Ze) about the neutral X-X axis (3)

3.4 The maximum bending moment of resistance if the maximum bending stress may NOT exceed 165 MPa (2)
[14]

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QUESTION 4: FRAMEWORK

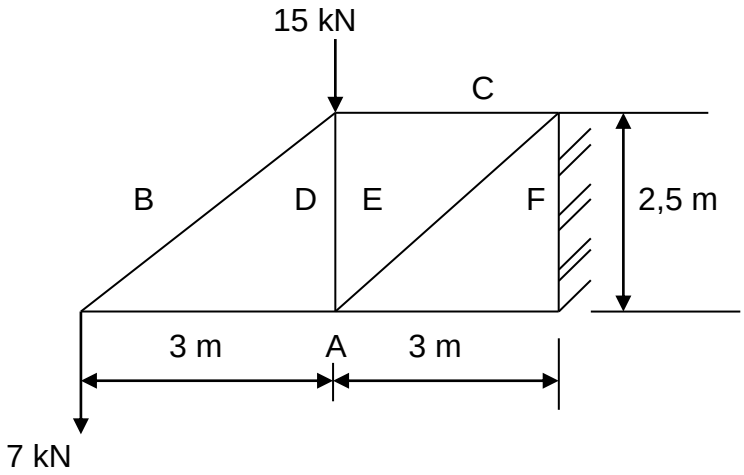


FIGURE 3

FIGURE 3 shows a loaded roof truss.

Determine the magnitude and nature of each member of the truss graphically if the roof is loaded as shown. (6)

NOTE: Tabulate the results as shown below.

MEMBER	MAGNITUDE	NATURE
BD		
DA		
CE		
ED		
EF		

(10)
[16]

TOTAL SECTION A: 54

SECTION B: DRAUGHTING

QUESTION 5: STEEL PROFILES

A concrete pad foundation is supporting a hollow square steel stanchion with a base plate bolted into the concrete. TWO gusset plates are welded on the outside of the stanchion with both gusset plates and the stanchion welded on the base plate.

Draw, to scale 1:5, a front view with the gusset plates and the TWO sides of the gusset plates in first-angle projection. Also draw the top view. Clearly show the constructional details.

Use the following specifications:

Pad foundation:	900 mm × 900 mm × 300 mm with FOUR M20 mm hook bolts fixed in both corners to hold the base plate
Base plate:	400 mm × 400 mm × 25 mm with Ø22 mm holes drilled 50 mm from the edges at both corners
Concrete base:	600 × 600 × 300 mm deep
Stanchion:	200 × 200 × 10 mm
Gusset plates:	400 × 200 × 8 mm
Holding down bolts:	M20 mm

Show at least 350 mm of the stanchion above the gusset plates. Insert the welding symbols. **[16]**

QUESTION 6: REINFORCED CONCRETE COLUMN

Draw, to scale 1:5, a vertical longitudinal section and a horizontal cross section through an octagonal reinforced concrete column with 300 mm sides. The column is reinforced with eight Y20 mm longitudinal bars and R8 mm helical binders at 150 mm centres. Show at least 500 mm of the length of the column and also show the position of reinforcement. (5 + 5) **[10]**

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QUESTION 7: CONCRETE L-BEAM

Draw a 254 × 254 × 89,2 kg/m H-profile steel column with parallel flanges supporting an edge beam.

Draw a 500 mm deep × 300 mm wide L-beam supporting a 200 mm thick reinforced concrete slab on top of the steel column. The beam is supporting a 270 mm cavity wall which has a plinth of 15 mm on the outside between the beam and the wall. The outer skin of the wall is face brick and the inner skin is plastered. The floor has a 20 mm granulite off finishing with 80 mm grano skirting. The beam and slab are cast monolithically.

Use the following specification:

Edge beam

Size:	600 mm × 300 m PLATR 19
Reinforcement:	4/Y20 mm tensile bars 2/R12 mm compression bars Stirrups of R8 mm at 200 mm c

Floor slab

Size:	200 mm thick
Reinforcement:	Main bars Y12 mm at 200 mm c/c Secondary bars Y10 mm at 300 c/c

Steel column

Size:	254 × 254 × 89,2 kg/m H-profile
Pressure plate:	254 × 254 × 10 mm placed on top of the column

Show at least about 1 500 mm of the length of the beam, a +/-500 mm part of the column and FOUR courses of brickwork.

[20]

TOTAL SECTION B: 46
GRAND TOTAL: 100

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(8060015)

FORMULA SHEET

Any applicable formula may also be used.

BM = $\frac{wl}{4}$

BM = $\frac{wl^2}{8}$

n = 5d
n = 5,5d

h = 9d
h = 11d

F = f.a

F = fs $\frac{\pi.D^2 n}{4}$

F = ft (W – n.d)

F = f_cD.t.n

F = $\frac{\pi.(\phi - 0{,}9382\rho)^2 n}{4}$

I = $\left[\frac{BD^3}{12}\right] + [2.area.y^2]$

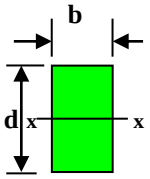
$\frac{M_r}{I} = \frac{f}{y} = \frac{E}{R}$

M = fZ

Z = $\frac{I_{NA}}{y}$

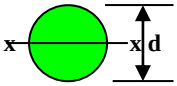
M = f$\frac{I}{y}$

M = $\frac{fbd^2}{6}$



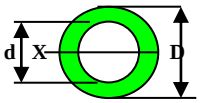
$I_{xx} = \frac{bd^3}{12}$

$Z_{xx} = \frac{bd^2}{6}$



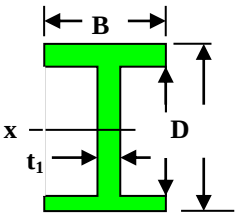
$I_{xx} = {}^1/_4 \pi r^4$

$Z_{xx} = \frac{\pi.d^3}{32}$

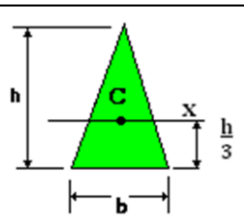


$I_{xx} = \frac{\pi}{64}(D^4 - d^4)$

$Z_{xx} = \frac{\frac{\pi}{64}(D^4 - d^4)}{D/2}$



$I_{xx} = \frac{BD^3}{12} - \frac{bd^3}{12}$



$x = \frac{bh^3}{36}$

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